

To what extent can deep learning enhance suspect profiling, and what ethical and methodological challenges limit its application?

1. Introduction

Along with the infamy of Jack the Ripper came the formal invention of suspect profiling. When police surgeon Thomas Bond compiled a criminal profile in hopes to aid the investigation to identify the infamous killer. Bond's analysis of autopsy reports and crime scene evidence marks the first recognized application of criminal profiling (Kocsis, 2004; Kapardis, 2024). While standardization of the practice would come in 1970s when the FBI established its Behavioral Science Unit (BSU) (Kapardis, 2024). Criminology has utilized suspect profiling and has been grounded in behavioral, psychological, and geographic analysis. While these methods have been criticized for subjectivity and bias (Petherick, 2020), such weaknesses not only undermine investigative reliability but also risk reinforcing stereotypes. The rise of deep learning (DL) methods now offer new possibilities for automating and enhancing profiling techniques. Existing research has highlighted the potential of DL in crime prediction and profiling tasks, though findings also stress both the methodological and ethical limitations (Mandalapu et al., 2023). This literature review asks: to what extent can deep learning enhance suspect profiling, and what ethical and methodological challenges limit its application? It critically examines methods and applications while highlighting the ethical risks around bias, privacy, and transparency.

2. Conceptual Background

Suspect profiling has been a systematic inference based on an offender's characteristics, crime scene evidence, victimology, and behavioral patterns. Early profiling drew heavily on psychological theories of its time and the early research of criminological typologies. The FBI's BSU created some of the most influential models in the approach of profiling (Kapardis, 2024). These approaches emphasized personality traits, demographic, and geographic profiling techniques, the goal of which was to predict an offender's home base from the distribution of crime. While this technique has been highly influential, it also relied on the intuition and subjective interpretation of experts, which has raised persistent concerns about reproducibility and scientific rigor (Petherick, 2020).

The limitations of profiling have promoted the growing computational and statistical approaches to suspect profiling. Early machine learning (ML) methods, such as logistic regression, decision trees, and support vector machines, were applied to crime prediction and offender classifications. These models offered a more data-driven approach with its ability to handle large numbers of criminal records, demographic attributes, and geospatial data (Bada and Nurse, 2021). The performance of this ML technique was however constrained and needed extensive feature engineering, and it showed difficulties generalizing across a diverse and complex dataset (Mandalapu et al., 2023).

Deep learning (DL) has been transformative in this context, as unlike its earlier models that depend on manually crafted features, DL architecture can automatically extract hierarchical representations from raw, unstructured inputs such as images, text, or

video. This has led to great performance improvements in fields such as facial recognition and natural language processing, both which directly relate to suspect profiling (Yadav and Vishwakarma, 2020). The improvement to DL also comes with new challenges, as models often function as “black boxes” making it difficult to explain decision pathways and their dependence on large datasets risks reproducing existing biases that are embedded in historical policing data (Almeida, McStay, and Sutcliffe, 2021). With this promise of DL scalability and enhanced predictive power, it also demands the adoption of critical reflection, ethical risks, and the tension between accuracy and accountability in suspect profiling.

3. Applications of Deep Learning in Suspect Profiling

3.1 Computer Vision Approaches

Computer vision is one of the most prominent applications of deep learning in suspect profiling. Particularly in facial recognition and behavioral biometrics. Convolutional Neural Networks (CNNs) have risen to a dominant way of extracting discriminative features from facial images, achieving high levels of accuracy on datasets in controlled environments (Yadav and Vishwakarma, 2020). Research has also extended to gait recognition, where DL models analyze surveillance footage to identify suspects by movement, even when faces are obscured. This can enable law enforcement agencies to automate the identity verification and suspect matching across large volumes of data.

While the testing has shown strong performance, significant reduction of accuracy has been shown when the models have been applied to uncontrolled, real-world settings done in various lighting, occlusion, and pose. These same studies have also shown that the facial recognition technologies (FRTs) can replicate and amplify demographic biases

present in their training datasets, which has produced a higher false positive rate for women and people of color (The Alan Turing Institute, 2020). This sort of disparity poses serious risks in suspect profiling, where misidentification can lead to wrongful accusations or discriminatory policing. Legal scholars have also warned that the reliance on FRT systems in criminal investigations raises challenges for due process and human rights, especially in the landscape of evolving regulations of the EU AI Act (Simmler, Sen, and Pesch, 2025).

Research has demonstrated that small, deliberate changes to input images can cause DNNs to misclassify individuals, which highlights how easily some vulnerabilities could be exploited within policing data (Almeida, McStay, and Sutcliffe, 2021). This further proves the tension between the technical potential of DL for visual suspect profiling and the risks associated with its deployment in high-risk environments. Taken together, while computer vision offers a powerful tool for automating visual identification, its reliability is contingent on the quality and representativeness of training data. Its adoption in suspect profiling is therefore not only a continued technical refinement issue but also a rigorous ethical and legal safeguard issue.

3.2 NLP/Text-Based Profiling

Another significant strand of research in DL-based suspect profiling has come from natural language processing (NLP). Offenders often leave behind a digital footprint that can be traced through online communication, manifestos, or social media interactions. This provides a rich textual dataset that can be digested for potential threat indicators. Deep learning models, particularly transformer-based architectures such as BERT and GPT, have demonstrated strong performance in text classification analysis. This

enables the detection of linguistic markers of criminal intent, extremist discourse, or deceptive communication (Bokolo, Chen, and Liu, 2023). Such systems could be employed to support early warning signs by flagging high-risk individuals or groups for further investigation.

Early NLP applications required intensive feature engineering and struggled with natural human language (Mandalapu et al., 2023). Now, DL models are excelling at capturing contextual nuance, irony, and semantic relationships, which are critical when analyzing social media data where communication is often very informal. Studies applying recurrent neural networks (RNNs) and long short-term memory (LSTM) architecture have shown notable improvements in profiling, cybercrime detection, and cyberbullying identification (Bada and Nurse, 2021).

However, the profound ethical and methodological concern the application of NLP in suspect profiling can be alarming. False positives remain a conservable risk, where sarcastic or metaphorical speech that can be misclassified as threats. Training data also often reflects sociocultural or political biases, meaning certain communities or linguistic groups may be disproportionately flagged. As Almeida, McStay and Sutcliffe (2021) argue, the uncritical adoption of text-based profiling can blur the boundary of legitimate law enforcement and unjustified monitoring of lawful expression. This is especially true in democratic countries where free speech protections may conflict with predictive security measures. Transformer models who achieve high performance, also suffer from highly opaque decision-making processes, this mirrors the broader “black box” problem in DL, where accuracy is gained at the expense of transparency.

3.3 Multi-Modal and Hybrid Profiling

Some of the most ambitious developments in deep learning-based suspect profiling combine multiple data modalities. Multi-modal approaches integrate visual, textual, and sometimes geospatial data, with the aim of producing a more comprehensive view of suspect behavior. Graph neural networks (GNNs) have been used to model criminal behavior to set individuals as nodes and their interactions as edges, which leads to the detection of hidden relationships and clusters of illicit activity (Mandalapu et al., 2023). These hybrid systems combine facial and linguistic recognition to generate richer offender profiles than the single-stream models could achieve. Proponents argue that such integration reflects the reality of contemporary investigations, where law enforcement must sift through diverse streams of digital evidence. Studies in crime prediction suggest that combining modalities are enhancing accuracy and robustness (Dakalbab et al, 2022).

Intensifying the ethical and methodological concerns, by aggregating disparate data sources, these systems magnify the scale of surveillance which continues to raise privacy implications. Almeida, McStay and Sutcliffe (2021) caution that while each modality may carry individual risks, their compounded integration shows intrusive monitoring. Hybrid models inherit and amplify the biases embedded in each input system, which may reinforce discriminatory outcomes. Another major critique concerns predictive policing, such as PredPol, which relies on historical crime data to forecast future criminal activity. These systems exemplify the dangers of multi-modal models trained on biased policing records, which can perpetuate cycles of over-policing in already marginalized communities. Without transparent auditing, multi-modal profiling risks replicating injustices and making them harder to contest due to DL opacity. While

these multi-modal can integrate complex evidence streams, these approaches highlight the urgent need for fairness-by-design methodologies and regulatory oversight before real-world deployment.

4. Challenges and Ethical Concerns

The application of deep learning has shown clear potential for advancing suspect profiling, but it also raises several challenges that extend beyond the technical. The most prominent issues around privacy and surveillance, model explainability, and legal frameworks that govern the use of AI in law enforcement.

One of the most consistent critiques is the susceptibility to replicate and amplify existing social biases. Facial recognition systems have been found to perform less accurately for women and ethnic minorities due to the unbalance in training data (The Alan Turing Institute, 2020). The risk of disproportionate false positives already falls on marginalized communities, and far from neutralizing human subjectivity, DL can entrench historical inequities in policing datasets (Almeida, McStay, and Sutcliffe, 2021).

The integration of DL into profiling also involves large-scale data collection. From CCTV footage to social media communications. This expansion of surveillance capabilities raises serious privacy concerns as individuals may be monitored and flagged without reasonable suspicions. Suspect profiling systems may therefore blur the line between legitimate investigation to mass surveillance, undermining civil liberties (Moreno, 2022).

The “black box” nature of DL models is another challenge for suspect profiling. The decision-making process of CNNs and transformers remains opaque. In a legal context, how can courts or defendants contest evidence derived from algorithms that cannot provide reasoning? This lack of transparency has the potential to erode trust and

compromise the principal of due process (Simmler, Sen, and Pesch, 2025). At a policy level, attempts to regulate AI profiling under GDPR, biometric and profiling data are classified as sensitive (European Parliament, 2020). The EU AI Act seeks to designate certain applications such as real-time facial recognition as “high-risk” to subject them to enhanced oversight. However, critics argue that regulation lags technological advancement and adoption, leaving questions of liability and accountability. Together, these challenges illustrate the effectiveness of DL in suspect profiling cannot be measured solely in accuracy and performance, but also in its ethical, legal, and societal considerations.

5. Research Gaps and Future Directions

Despite promising advances, the literature on deep learning in suspect profiling remains fragmented with notable gaps in methodologies and ethics. While explainable AI (AXI) methods such as LIME and SHAP have been explored in other domains, it remains unseen in forensic applications. Without interpretable outputs, DL profiling risk undermining legal accountability and due process (Simmler, Sen, and Pesch, 2025). Another gap concerns the diversity in data. Most profiling systems are trained in Western-centric data demographics, which risks embedding systemic biases when applied on the global level. Future work should develop more inclusive datasets that show a wider range of social, cultural, and demographic contexts (Mandalapu et al., 2023).

Interdisciplinary collaboration also needing further development, as most of the technical literature evaluates models solely on predictive accuracy. This is neglecting the legal, ethical, and criminological perspective. Almeida, McStay, and Sutcliffe (2021)

argue that meaningful safeguards require the joint effort of AI researchers, policymakers, and social scientists to become successful. This balance in design of DL systems would help balance innovation with accountability. Finally, future directions should include fairness-by-design approaches. Where safeguards against bias and privacy violations are created in the model-development stage to proactively work to mitigate risks before deployment.

6. Conclusion

This review has traced the evolution of suspect profiling from its historical roots in psychological and behavioral analysis to the contemporary application of deep learning techniques. It has shown the approaches in improving the process of profiling, showing how CNNs, transformers, and hybrid systems have achieved accuracy and automation. Simultaneously, this review emphasized the limitations of these models, particularly in their susceptibility to bias, opacity, and ethical concerns. A critical finding across the literature is that deep learning does not eliminate the challenges of bias that plagued traditional profiling. It risks embedding them into algorithmic systems that are less transparent and harder to contest. Future progress depends on the development of explainable, fair, and legally accountable models designed in collaboration with social scientists and policymakers. Only then can deep learning contribute meaningfully to justice rather than reproducing existing inequities.

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